

Chromosome-level genome assembly of the *Erythrina* gall wasp, *Quadrastichus erythrinae* (Hymenoptera: Eulophidae)

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The *Erythrina* gall wasp, *Quadrastichus erythrinae*, is an invasive gall-inducing chalcidoid wasp and a major pest of the endemic wiliwili tree (*Erythrina sandwicensis*) in Hawai'i. As a foundation for associated research, we generated a chromosome-level genome assembly from a wild-collected female measuring <2 mm. The final assembly consists of 5 scaffolds representing the 5 autosomes totaling 399 Mb (N50 = 75.6 Mb) and 1 unplaced 16 kb contig. BUSCO analysis recovers 91.1% of conserved Hymenoptera orthologs, representing the first chromosome-scale genome for the genus *Quadrastichus*. Comparative genomic analyses reveal syntenic conservation across Hymenoptera despite deep evolutionary divergence, with strongest collinearity to the chalcidoid *Nasonia vitripennis*. Genome size variation is largely explained by repeat content, and *Q. erythrinae* exhibits high proportions of unclassified transposable elements similar to the cynipid gall inducer. We also assembled a complete genome of its endosymbiont, *Wolbachia pipientis*. Together, these genomic resources provide a foundation for comparative, evolutionary, and applied research aimed at managing this invasive pest.

Keywords: Tetrastichinae; Chalcidoidea; secondary phytophagy; insect pest; genome assembly; *Wolbachia*

Introduction

The tetrastichine eulophids (Chalcidoidea: Eulophidae, Tetrastichinae) are one of the largest subfamilies of chalcidoid wasps, comprising over 100 genera and approximately 3,000 known species (UCD Community 2023). This hyperdiverse but poorly defined group mostly consists of parasitoids but also includes 16 genera with recorded phytophagy and the ability to induce galls on both monocot and dicot plants (Zhang et al. 2022). *Quadrastichus* is one of the larger genera within the “*Tetrastichus* group” sensu (Rasplus et al. 2020), with 95 described species and many others still undescribed (UCD Community 2023; Kärnäs et al. 2025).

The *Erythrina* gall wasp, *Quadrastichus erythrinae*, is a gall-inducing species first identified in 2004 when it emerged as a serious invasive pest (Kim et al. 2004). Despite being less than 2 mm in size, the wasp induces galls on the young leaves, stems, and petioles of *Erythrina* (Fabaceae: Faboideae) species (Fig. 1), often leading to tree decline and death (Kim et al. 2004; Lin et al. 2021). Likely originating in East Africa, *Q. erythrinae* has spread rapidly across tropical and subtropical regions of Asia, Oceania, and the Americas (Kim et al. 2004; Lin et al. 2021). It causes significant damage to both endemic and introduced *Erythrina* species. *Quadrastichus erythrinae* posed a major threat to *Erythrina*

sandwicensis (wiliwili), the only endemic *Erythrina* species on the Hawaiian Islands. This tree holds great societal and cultural value for native Hawaiians and is of significant ecological importance as one of the few deciduous trees in the endangered lowland dry forests (Kaufman et al. 2020). The wiliwili was near the brink of extinction due to *Q. erythrinae* infestations, until the successful deployment of the biological control agent *Eurytoma erythrinae* reduced infestation levels (Kaufman et al. 2020).

We sequenced and assembled a chromosome-level genome for *Q. erythrinae*, leveraging its raw sequencing data to also recover the complete genomes of its *Wolbachia* endosymbionts. This approach demonstrates how mining host genome data can effectively characterize cobionts, providing crucial ecological context for their hosts.

Materials and methods

Source material

Galls from the leaves of *E. sandwicensis* were collected near Wawaloli Beach Park in Kaiminani, HI (19.7145, −156.0491) on 3.V.2025 and returned to the United States Daniel K Inouye U.S. Pacific Basin Agricultural Research Center in Hilo, Hawai'i, United States. Adults were reared out of the galls, identified,

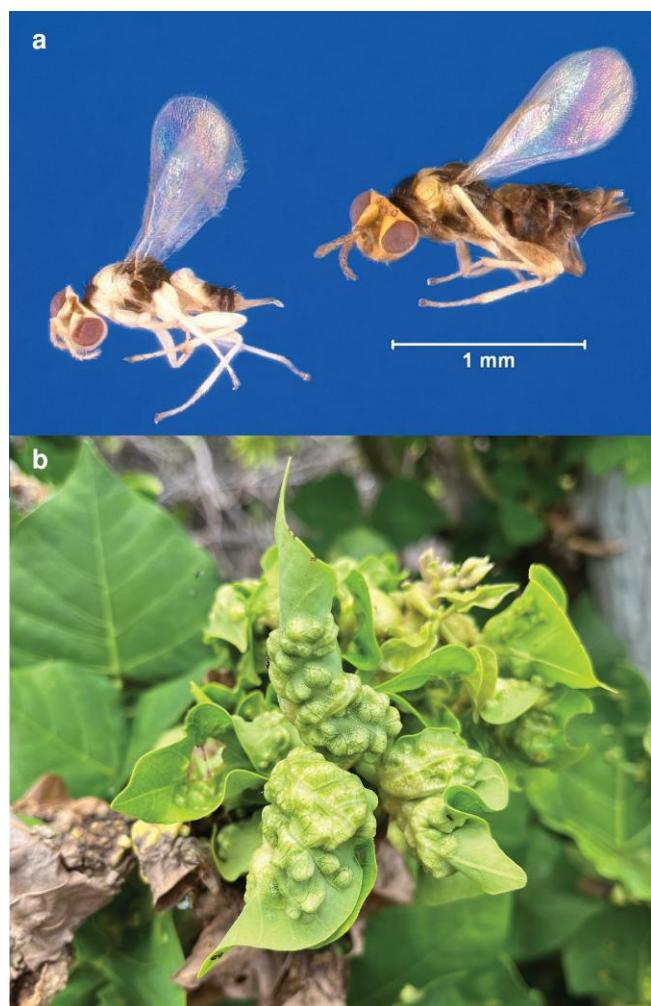


Fig. 1. *Quadrastichus erythrinae*. a) Lateral habitus of male (left) and female (right) adult *Q. erythrinae*. Photo by Erich G. Vallery, USDA Forest Service—SRS-4552, Bugwood.org, licensed under CC BY 3.0. b) Galls induced by *Q. erythrinae* on wiliwili tree in Kona, Hawai'i.

flash-frozen alive using liquid nitrogen, and stored at -80°C until genomic library preparation.

Library preparation and sequencing

The whole body of a single female wasp was homogenized into a fine powder while kept frozen using a SPEX SamplePrep 2010 Geno/Grinder (Cole Parmer, Metuchen, New Jersey, United States) and underwent high molecular weight (HMW) DNA extraction using the fresh or frozen tissue protocol of the Qiagen MagAttract HMW DNA Kit (Qiagen, Hilden, Germany). Following isolation, the HMW DNA (92.5 ng) was sheared to a mean fragment length of 20 Kb with a Megaruptor 3 (Diagenode, Denville, New Jersey, United States) and prepared into a PacBio SMRTBell library using the SMRTBell Express Template Prep Kit. 3.0 (Pacific Biosciences, Menlo Park, California, United States) using a bar-coded adapter. After DNA isolation, shearing, and library preparation, the sample was purified using solid-phase reversible immobilization beads (SPRI beads) (DeAngelis et al. 1995) and quantified using fluorometry and spectrophotometric absorbance ratios (DeNovix Inc., Wilmington, Delaware, United States). Fragment length distribution after each step was determined by Femto Pulse or Fragment Analyzer (Agilent Technologies, Santa

Clara, California, United States). The resulting library was pooled with other samples and sequenced on a PacBio Revio system using a 30-h movie length on 1/10th of a Revio SMRTCell. Raw subreads were converted to HiFi data using the PacBio SMRTLink software v.10.1.

Concurrent to HiFi sequencing, a pool of twelve males was used to prepare an enriched chromosome conformation capture (HiC) library. Tissues were homogenized in $1\times$ phosphate-buffered saline, and nuclei were crosslinked in a 2% formaldehyde solution. Following crosslinking, the sample was lysed and digested using the restriction enzymes DdeI and DpnII. To enrich the sample for proximity ligated fragments, a biotin-labeled dATP fill-in step was performed prior to proximity ligation so fragments could be captured downstream. Following proximity ligation, a crosslink reversal step was performed followed by 2 DNA purification steps using SPRI beads, the removal of biotin from unligated ends, and another DNA purification step. The sample was size-selected using SPRI beads, biotinylated ligation products were captured, and the sample was prepared into a short-read sequencing library using the NEBNext Ultra II DNA Library Prep Kit (New England Biolabs, Ipswich, Massachusetts, United States). The final libraries were sequenced on a partial flowcell using the AVITI 2×150 Sequencing Kit Cloudbreak FS High Output kit on the Element AVITI System (Element Biosciences, San Diego, CA). Following sequencing, raw reads were base called using bases2fastq v.2.3.0.2116803307.

Genome assembly, assessment, and contaminant removal

HiFi reads were screened and filtered for adapter-contaminated sequence artifacts using FCS-Adaptor and HiFiAdapterFilt (Sim et al. 2022) (<https://github.com/ncbi/fcs>). The resulting HiFi reads were used to assemble contigs using HiFiASM (v.0.24.0-r702) (Cheng et al. 2021, 2022). The contig assemblies were subsequently purged of duplicate contigs using PurgeDups (Guan et al. 2020), and the duplicate-purged contig assembly served as the reference to map HiC reads using BWAmem 2 (v.2.2.1) (Vasimuddin et al 2019). The resulting mapped reads were filtered for artifact PCR duplicates using Picard (v.3.2.0) (Picard2019toolkit, 2019 <https://github.com/broadinstitute/picard>). A contact map was generated from the de-duplicated mapped reads using the YaHS pipeline (Zhou et al. 2023). Visualization of the contact map and minor manual editing were achieved using Juicebox (v.2.15) (Durand et al. 2016). Minimap2 (v.2.22-r1101) was used to map HiFi reads back to the contig assembly and calculate coverage of each contig, the $-\text{auto}$ function and $-\text{genome}$ mode of BUSCO (v.5.8.3; Manni et al. 2021) was used to select the appropriate taxon database and estimate genome completeness, and BLAST+ and Diamond were used to perform nucleotide alignments to the NCBI nucleotide database (accessed November 2025) and UniProt protein database (accessed November 2025), respectively (Camacho et al. 2009; Li, 2018; Buchfink et al. 2021, Tegenfeldt et al. 2025). The resulting outputs of minimap2, BUSCO, BLAST+, and Diamond were summarized and visualized using Blobtools2 and blobblurb (Challis et al. 2020; <https://github.com/sheinasim/blobblurb>). Additional taxonomic assignments of contigs and contig fragments were performed using the FCS-GX (Astashyn et al. 2024). The confluence of the taxonomic assignments based on nucleotide alignment, protein alignment, and FCS-GX was used to identify contigs assigned to *Wolbachia* and remove nonarthropod contigs from the assembly. The primary and alternate assemblies were submitted to the

National Center for Biotechnology Information (NCBI), along with the *Wolbachia* sequence, with the accession SAMN57203050.

RNA extraction and sequencing

We sequenced RNA to aid in genome annotation. Specifically, total RNA was extracted from 3 pools of 4 male specimens separated by body parts (head, thorax/mesosoma, and abdomen/metasoma), through homogenization of snap frozen tissue in Tri Reagents and then extraction using the Zymo Direct-zol Magbead Total RNA kit (Zymo Research, Irvine, California, United States) on a KingFisher Flex 96 system (Thermo Fisher Scientific, Waltham, Massachusetts, United States). From total RNA, a poly(A) cDNA library was generated using the NEB Ultra II RNA Library Prep Kit (New England Biolabs, Ipswich, Massachusetts, United States) using oligo(dT) selection. The final RNA-seq libraries were sequenced on a partial flowcell using the AVITI 2 × 150 Sequencing Kit Cloudbreak FS High Output kit on the Element AVITI System (Element Biosciences, San Diego, California, United States). Following sequencing, raw reads were base called using bases2fastq v.2.3.0.2116803307 the same way as HiC above.

Genome annotation

We performed genome annotation of *Q. erythrinae* using the Eukaryotic Genome Annotation Pipeline—External (EGAPx) (<https://github.com/ncbi/egapx>) v0.4.0. Short-read RNA-seq from a pool of 4 whole male bodies was used as transcriptome evidence and aligned to the respective reference using STAR. Miniprot was used to align Hymenoptera protein sequences to the reference. Gnomon (https://www.ncbi.nlm.nih.gov/refseq/annotation_euk/gnomon/) was used for gene prediction using protein and RNA-seq alignments and ab initio predictions based on HMM. Lastly, EGAPx adds functional annotations to the final structural annotation set based on orthology and model type and quality. The *Wolbachia* genome was annotated using Prokka v1.14.6 (Seemann 2014).

Mitochondrial genome

We identified all potential mitochondrial genome contigs using the MitoHiFi pipeline (Uliano-Silva et al. 2023). MitoHiFi implemented a BLAST search for contigs that have a high similarity to the mitochondrial genome of *Nasonia vitripennis* (NCBI RefSeq accession NC_066201.1) (Camacho et al. 2009) and selected the contig with the greatest similarity. Mitochondrial genes were then structurally annotated using intervals from the same mitochondrial genome used in the BLAST search through the MitFi annotation program in MITOS2 (Bernt et al. 2013). A representative mitochondrial genome was submitted to NCBI under the accession PX622211.

Synteny across hymenoptera

Synteny conservation among 8 chromosome-scale diverse hymenopteran assemblies (Table 2) was used to place the *Q. erythrinae* genome in a comparative genomic context. Analyses were performed using all annotated genes by first extracting the longest isoform per gene from the RefSeq GFF annotations using AGAT (v1.4.3) (Dainat et al. 2025), with corresponding nucleotide CDS sequences inferred with TransDecoder (v5.7.1) (Haas et al. 2013). Synteny inference was performed in JCVI (v1.4.16) (Tang et al. 2024) using LAST (v2.37.4) (Frith et al. 2010) for pairwise alignment, applying a cscore threshold of 0.99 to retain reciprocal best-hit gene pairs and filtering syntenic blocks containing at least

10 gene anchors. Synteny was visualized using pairwise dot plots and whole-chromosome karyoplots. Note that JCVI visualizations are based solely on gene order and anchor counts, so chromosome lengths in JCVI plots do not reflect physical chromosome sizes.

We corroborated this analysis with BUSCO-defined genomic anchors and chromsyn (v1.6.1) (Edwards et al. 2022), which infers collinear regions from the relative order and spacing of conserved single-copy orthologs. All genomes were annotated with BUSCO (v5.8.3) using the Hymenoptera lineage dataset (hymenoptera_odb12). Conserved regions were defined by clustering BUSCO loci using chromsyn, retaining only syntenic blocks larger than 100 Kb which contain more than 2 BUSCO genes to emphasize macro-syntenic structure. Shared synteny among assemblies was quantified by counting these pairwise BUSCO-defined blocks with visualization using log-scaled heatmaps in R (v4.4.1) and the tidyverse (v2.0.0) (R Core Team 2017; Wickham et al. 2019).

Repeat landscape

We quantified total repetitive DNA content for the same 8 chromosome-level hymenopteran assemblies using EarlGrey (v6.3.5) (Baril et al. 2024). Repeat annotation summaries were derived from GFF files using custom R scripts, assigning each genomic position to a single repeat subclass using a fixed hierarchical order (LTR, LINE, Penelope, SINE, DNA, Rolling Circle, Other, Unclassified, Non-Repeat), with overlapping intervals resolved by priority-based merging in this order using R, tidyverse, and GenomicRanges (v1.56.2) (Lawrence et al. 2013). From these repeat annotations, we extracted 3 summary metrics per transposable element (TE) subclass: (i) total cumulative genomic coverage (Mb), (ii) the number of distinct TE families, and (iii) the proportion of the genome covered by each subclass. To assess the relationship between repeat abundance and genome size while accounting for shared evolutionary history, we performed phylogenetic generalized least squares (PGLS) regression using caper (v1.0.4) (Orme et al. 2025) in R. Both genome size and total repeat genomic coverage (Mb) were log-transformed, and Pagel's λ was estimated by maximum likelihood to identify phylogenetic signal. As a non-parametric complement, we computed Spearman's rank correlation between genome size and repeat genomic coverage (Mb). Both analyses were repeated excluding *Belonocnema kinseyi* as its substantially larger genome size relative to the other assemblies could disproportionately influence results.

Results and discussion

Genome assembly metrics

PacBio HiFi sequencing yielded a highly contiguous genome for *Q. erythrinae*. From 1 Revio cell where the barcoded *Q. erythrinae* SMRTBell library represents 1/10th of the sequencing run, 621,619 HiFi reads (9.5 Gb of HiFi data) were obtained, of which 18 (0.00002% of reads) were discarded after filtering with HiFiAdapterFilt. K-mer analysis of the final assembly relative to the PacBio HiFi reads used to create the contig assemblies reported a raw QV score of 55.031. The genome coverage was estimated at 22.6× coverage using GenomeScope2. The initial HiFiASM assembly consisted of 41 contigs, L50 of 5, N50 of 32.9 MB, L90 of 5, N90 of 4.2 MB, and a genome size of 399.197 MB (Table 1). The final assembly consists of 41 contigs representing 5 autosomes, a mitochondrial genome, and an unplaced 16.7 kb contig composed completely of repeat elements as identified by EarlGrey. Scaffolding with HiC further improved the contiguity of the genome, while BlobToolKit identified 1 scaffold of nonarthropod origin, mapped to the endosymbiotic

bacterium *Wolbachia pipientis*. Upon removal of the endosymbiont, the final genome consisted of 6 scaffolds totaling 399.201 MB, an N50 of 75.621 MB, L50 of 3, N90 of 59.692 MB, and L90 of 5 (Fig. 2a). The assembly has a fairly complete Hymenoptera BUSCO, with 91.1% single-copy complete, 2.4% fragmented, 1.23% duplicated, and 8.9% missing using hymenoptera_odb12 ($n = 5,920$) (Fig. 2a). This genome is in parity with other annotated Chalcidoidea genomes in NCBI (eg *N. vitripennis* 95.3%, *Copidosoma floridanum* 90.6%, *Phymastichus coffea* 92.7%, *Trichogramma pretiosum* 92.7%). HiC scaffolding revealed 5 chromosomes (Fig. 2b). Repetitive elements encompassed 57.2% (229.8 MB) of the *Q. erythrinae* genome. Among these, TEs class I (LTRs, LINEs, and Penelope-like elements) represented 19.2% of the genome, while TEs class II (DNA transposons and Rolling Circle helitrons) represented 11.0%. Additionally, 24.7% of the TEs were unclassified, while other repeats (simple repeats, microsatellites, RNAs) consisted of 2.4%. The *Q. erythrinae* genome has many quality metrics that meet or exceed the standards of the Earth BioGenome Project, such as a contig N50 well over 1 MB (N50 = 40.5 MB) and single-copy complete BUSCOs over 90% (Lawniczak et al. 2022). Additionally, the size of the scaffold genome (399 MB) is well within the range of size of available genomes on NCBI for Eulophidae (172–509 MB) and Chalcidoidea (172–1,085 MB).

The mitochondrial genome of *Q. erythrinae* is 14,607 bp in length and contains the full complement of 13 protein-coding genes, 22 tRNA genes, and both rRNA genes. The arrangement of genes is *cox1-cox2-atp8-atp6-cox3-nad3-nad2-rns-rnl-nad1-cytb-nad6-nad4L-nad4-nad5*, which differs from the type 5 and type 6 arrangements found in many chalcidoids, including other *Quadrastichus* species (Zhu et al. 2023). In particular, the *cox/atp/nad3* block is inverted compared to type 5/6, with type 5/6 showing *nad3-cox3-atp6-atp8-cox2-cox1* while the genome of *Q. erythrinae* has *cox1-cox2-atp8-atp6-cox3-nad3*; the *nad5-nad4-nad4L* block is also reversed relative to type 5/6, and *nad2* is moved

from the last position in type 5/6 to an early position in the genome, reflecting extensive mitochondrial gene rearrangement.

The *Wolbachia* genome wQua is 1,346,620 bp, with a BUSCO score of 98.0% single-copy complete, 0.3% fragmented, 1.7% duplicated, and 0% missing using rickettsiales_odb12 ($n = 345$). It

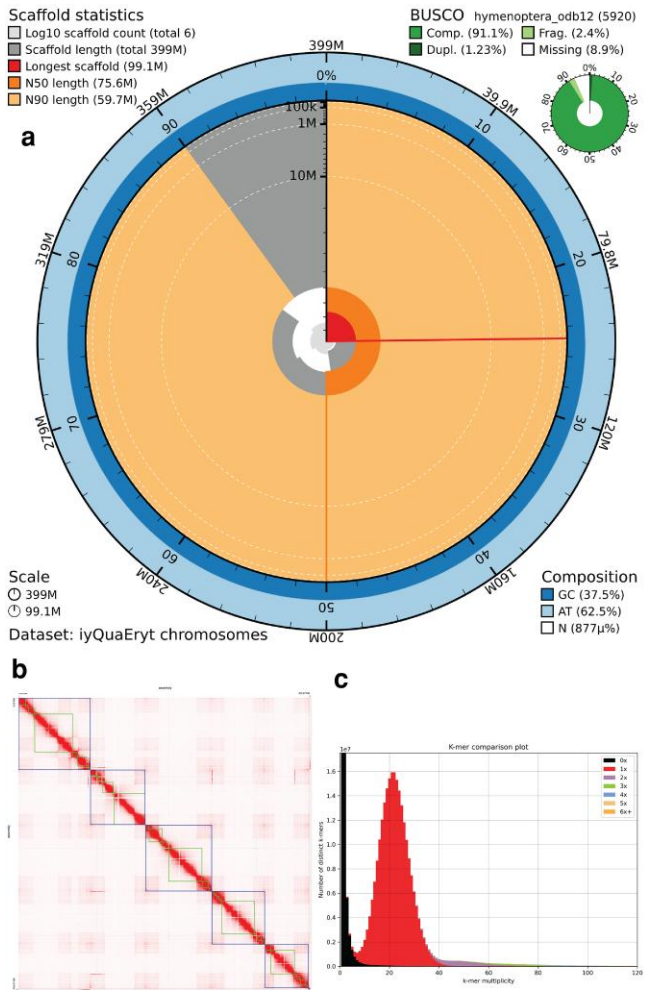


Fig. 2. Statistics of the *Q. erythrinae* assembly. a) Snail plot visualization of the scaffold assembly with BUSCO assessment using Hymenoptera conserved orthologs odb12. b) HiC contact map indicates that contigs can be grouped into 5 major scaffolds. c) K-mer spectra plot showing the representation graph of k-mers in the assembly relative to the raw HiFi data.

Table 1. Assembly statistics of the *Q. erythrinae* genome contig and scaffold assemblies.

	Contig	Scaffold
Total fragments	41	6
N50 (MB)	32.9	75.6
L50	5	3
N90 (MB)	4.2	59.7
L90	5	3
Total (MB)	399.2	399.2

Table 2. Assembly statistics and accession information for the 8 hymenopteran genomes analyzed in this study.

Species	Higher groupings	Scaffold N50 (Mb)	Scaffolds	Size Gb	Accession
<i>Q. erythrinae</i>	Chalcidoidea, Proctotrupomorpha, Parasitica	75.62	6	0.40	This study
<i>N. vitripennis</i>	Chalcidoidea, Proctotrupomorpha, Parasitica	24.76	436	0.30	GCF_009193385.2 (Dalla Benetta et al. 2020)
<i>B. kinseyi</i>	Cynipoidea, Proctotrupomorpha, Parasitica	150.97	5,520	1.54	GCF_010883055.1
<i>D. longicaudata</i>	Ichneumonoidea, Parasitica	7.96	246	0.19	GCF_034640455.1
<i>A. mellifera</i>	Apoidea, Aculeata	13.62	177	0.23	GCF_003254395.2 (Wallberg et al. 2019)
<i>Vespa crabro</i>	Vespoidea, Aculeata	9.77	99	0.23	GCF_910589235.1 (Crowley et al. 2022)
<i>S. invicta</i>	Formicoidea, Aculeata	26.23	219	0.38	GCF_016802725.1 (Helleu et al. 2022)
<i>Neodiprion pinetum</i>	Tenthredinoidea, Symphyta	41.40	112	0.27	GCF_021155775.2

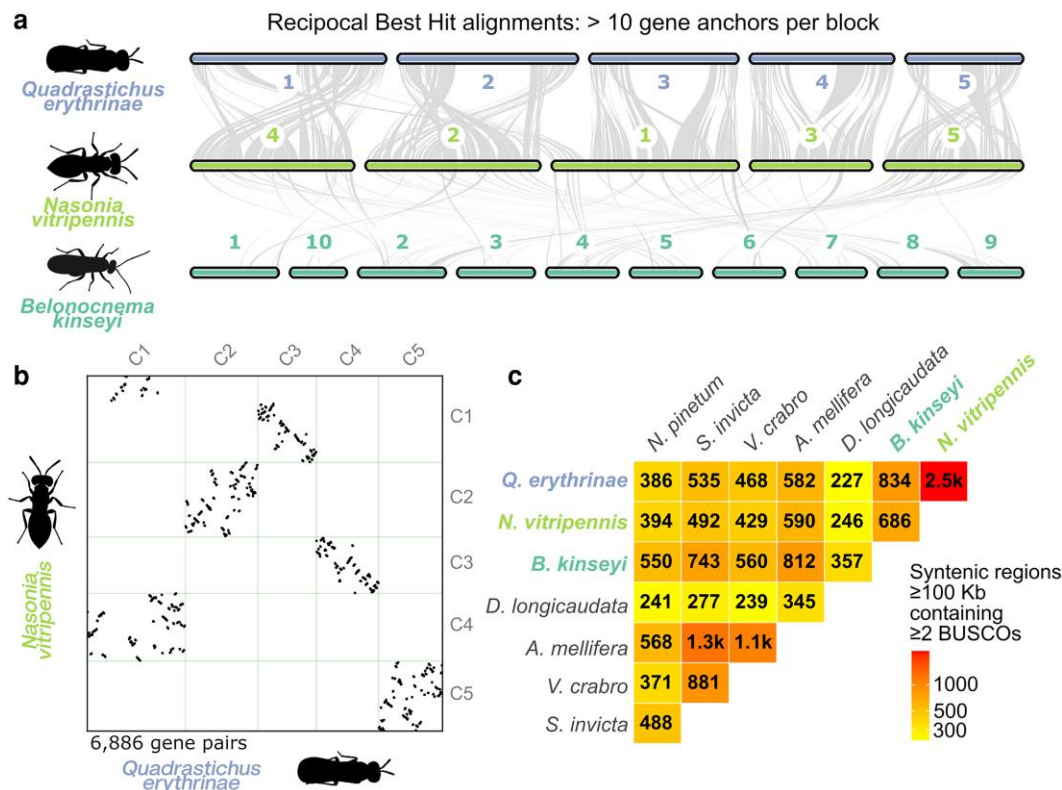


Fig. 3. Gene-based synteny across 8 hymenopteran genomes. a) JCVI karyoplot showing reciprocal best-hit gene alignments across chromosomes inferred from the full genome annotation. Chromosome lengths reflect gene order only and do not represent physical sizes. b) Pairwise dot plot showing collinearity between *Q. erythrinae* and *N. vitripennis* based on reciprocal best-hit alignments. c) Pairwise counts of conserved macro-syntenic blocks inferred with chromsyn using BUSCO-defined anchors, filtered for collinear blocks ≥ 100 Kb containing ≥ 2 genes.

belongs to Supergroup A, which is commonly found among terrestrial insects including Hymenoptera (Vancaester and Blaxter 2023).

Chromosome-level syntenic conservation

Whole-genome alignments indicate high syntenic conservation between *Q. erythrinae* and other hymenopteran genomes despite considerable variation in haploid chromosome counts and assembly contiguity across taxa (Fig. 3). Reciprocal best-hit comparisons across all annotated genes reveal the highest degree of orthology between the 2 chalcidoid species *Q. erythrinae* and *N. vitripennis* ($n=7,144$), intermediate counts in other comparisons, and the lowest between *B. kinseyi* and *D. longicaudata* ($n=691$) after filtering for syntenic blocks containing ≥ 10 gene anchors (Fig. 3a). Gene order between *Q. erythrinae* and *N. vitripennis* is largely collinear (Fig. 3b), consistent with extensive conservation of chromosomal architecture within Chalcidoidea.

At the level of deeply conserved genes, BUSCO counts on chromosome-scale scaffolds differed modestly between Proctotrupomorpha (*Q. erythrinae*, *N. vitripennis*, *B. kinseyi*) (mean = 5,365, range = 5,234 to 5,543) and other lineages (mean = 5,795, range = 5,752 to 5,826), with overlapping ranges and only a marginally significant difference in mean values (2-sample t-test P -value = 0.04). This pattern indicates broadly comparable recovery of deeply conserved single-copy orthologs across assemblies, consistent with BUSCO scores as a measure of expected gene content completeness rather than a sensitive indicator of lineage-specific gene gain or loss (Waterhouse et al. 2018). Even across deep evolutionary distances in Hymenoptera (~250-My divergence; Blaimer et al. 2023), we recover numerous macro-syntenic blocks ≥ 100 Kb containing at least 2 BUSCO genes, reflecting

persistence of large conserved genomic segments despite lineage-specific rearrangements (Fig. 3c). The greatest number of conserved BUSCO-anchored syntenic blocks is between the 2 chalcidoids *Q. erythrinae* and *N. vitripennis* ($n=2,482$), whereas the fewest occur between *Q. erythrinae* and the braconid *Diachasmimorpha longicaudata* ($n=227$). This pattern reflects known heterogeneity in chromosome number evolution across Hymenoptera, with broad karyotypic diversity involving frequent rearrangements accompanying speciation and life-history shifts (Gokhman 2022).

Repeat evolution

Repeat composition varied substantially across the sampled hymenopteran genomes (Fig. 4). Notably, no short interspersed nuclear elements (SINEs) were detected on chromosome-scale scaffolds in any assembly, consistent with previous reports that SINEs comprise a negligible fraction (<1%) of TE content in Hymenoptera relative to other insect orders (Petersen et al. 2019). Despite relatively similar gene counts across species (ranging from 33,367 in *Apis mellifera* to 44,526 in *Solenopsis invicta*), genome size varied substantially, leading us to test whether differences in repeat abundance explain this interspecific variation. Across the 7 species excluding the exceptionally large-genome outlier *B. kinseyi* (1.54 Gb, Table 2), phylogenetically corrected regression revealed that total repeat content is a strong predictor of genome size ($\beta=0.32 \pm 0.086$, $P=0.014$, $\lambda=0.0$), explaining 73.6% of variance in genome size (Fig. 4d). A nonparametric Spearman correlation corroborated this association ($\rho=0.89$, $P=0.012$), further supporting a positive relationship between repeats and genome size (Fig. 4d). These associations remained significant with inclusion of *B. kinseyi* (PGLS $P<0.01$; Spearman's $\rho=0.93$;

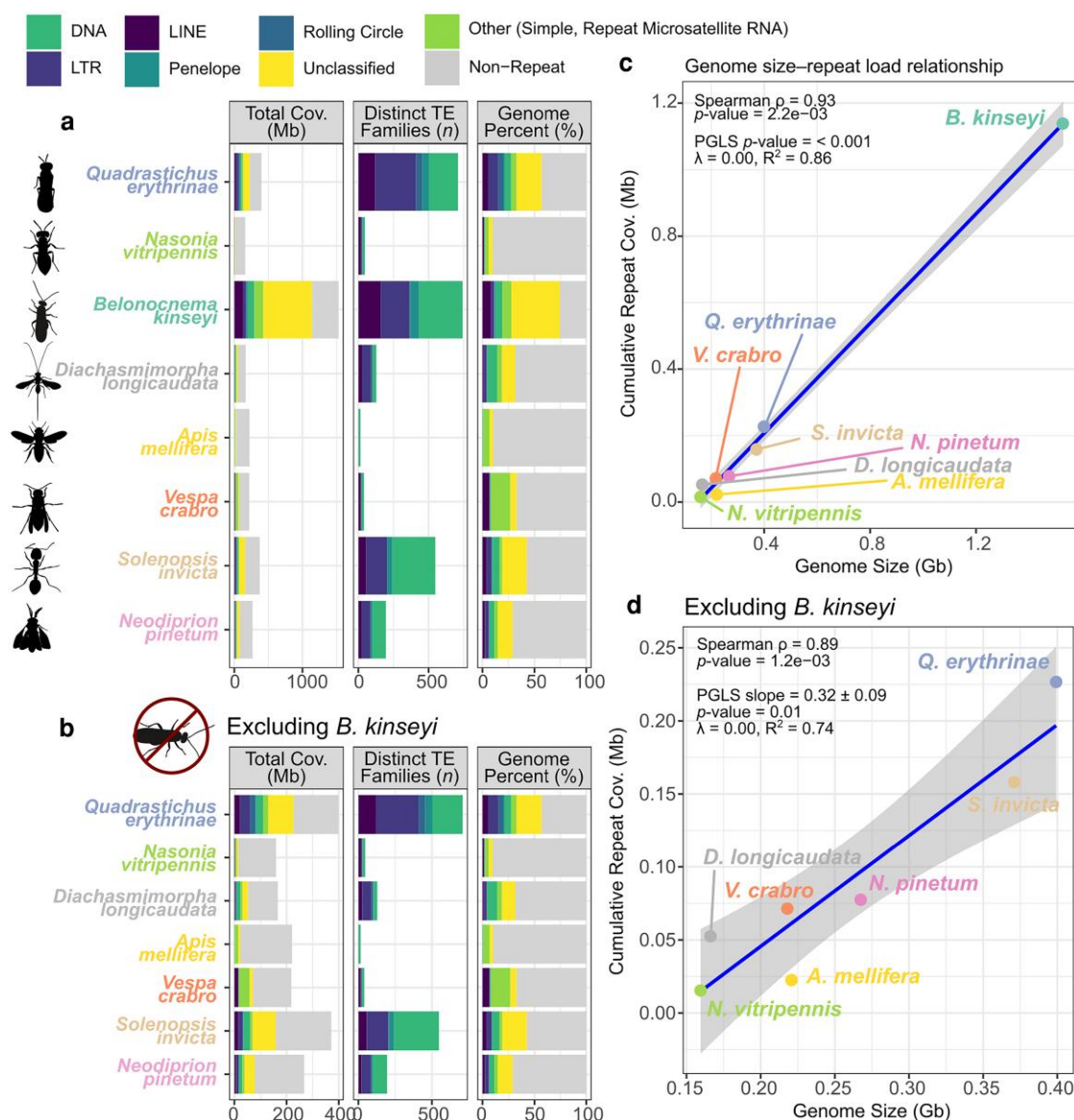


Fig. 4. Repeat landscape across hymenopteran genomes. a) Total repetitive DNA content was quantified for 8 chromosome-level hymenopteran assemblies using EarlGrey and summarized by repeat subclass. Facets show total cumulative genomic coverage (Mb), number of distinct TE families, and percent of the genome occupied by each repeat subclass. “Non-Repeat,” “Unclassified,” and “Other” categories were excluded from TE family count visualizations. (b) Same as above, except excluding the large-genome outlier *B. kinseyi* (bottom). c) Association between genome size (Gb) and cumulative repeat coverage (Gb) with PGLS regression and Spearman correlation results annotated. d) Same analysis excluding *B. kinseyi*.

Fig. 4c). These patterns align with patterns observed broadly across insects, where repetitive sequence content often underlies genome size variation (Cong et al. 2022; Cook et al. 2025) and contributes to the C-value enigma by driving genome size expansions independent of coding content (Gregory 2001). Interestingly, despite *Q. erythrinae* having a much smaller genome than *B. kinseyi*, both distantly related gall inducers harbor large proportions of unclassified TEs, accounting for 24.7% and 52.8% of their genomes, respectively (Fig. 4a). This contrasts with the more closely related chalcidoid parasitoid *N. vitripennis*, which also has a small genome but a much lower proportion of unclassified TEs (13.5%). Although it is premature to infer convergent genomic patterns among phytophagous lineages within Proctotrupomorpha, future characterization of these unclassified TEs across broader taxonomic sampling within the group will be important for determining whether genome expansion is associated with the transition to

phytophagy. We note that *S. invicta* also exhibits an increase in distinct TE families (Fig. 4a), highlighting that lineage-specific TE proliferation occurs across Hymenoptera and emphasizing the need for broader sampling to disentangle ecological from lineage-specific effects.

Conclusion

The *Q. erythrinae* reference genome, generated from a single female measuring <2 mm with <100 ng of input DNA for library preparation and occupying approximately one-tenth of a PacBio HiFi SMRT Cell, represents a major advance as the first genome for a phytophagous eulophid and the first chromosome-scale assembly for the genus *Quadrastichus*. Importantly, this demonstrates that high-quality, chromosome-level assemblies can be obtained from extremely small insects without whole-genome

amplification, defining a practical lower bound for direct long-read sequencing of minute taxa. Given current PacBio Revio pricing, the HiFi sequencing for this genome corresponds to only ~\$100 USD in marginal sequencing cost, underscoring the increasing feasibility of low-cost de novo genome assembly for tiny hymenopterans, even while acknowledging additional costs associated with sample preparation and library construction. As only the fifth genome available for the hyperdiverse subfamily Tetrastichinae and only the second at chromosome scale, this resource expands genomic representation for Chalcidoidea. Comparative analyses reveal strong macro-syntenic conservation across Hymenoptera despite extensive karyotypic variation, highlighting the utility of this assembly for evolutionary and phylogenomic studies. Patterns of repeat content further indicate that genome size evolution in Hymenoptera is largely driven by repetitive elements, with *Q. erythrinae* exhibiting relatively low repeat divergence compared to other lineages. This high-quality assembly facilitates population genomic, diagnostic, and phylogenetic analyses, including the identification of loci underlying cryptic species boundaries and traits such as gall induction. In addition, recovery of a complete *Wolbachia* genome directly from host sequencing data underscores the value of chromosome-level assemblies for integrative host-symbiont research. Together, these resources establish the *Q. erythrinae* genome as a foundational genomic model for both applied pest management and comparative hymenopteran genomics.

Data availability

Custom scripts, plotting input files, and alignments used for chromosome-scale synteny and repeat analyses are available at https://github.com/merondun/hymenopteran_alignments and are permanently archived on Zenodo (DOI: <https://doi.org/10.5281/zenodo.18157483>). The primary and alternate genome assemblies are hosted at the NCBI under BioProject Accessions PRJNA1404342 and PRJNA1404344, respectively. The sample used for the contig assembly is described under BioSample SAMN54712453 and registered under the Darwin Tree of Life ID iyQuaEryt1. Raw read data were submitted under SRA accessions: SRR36875831-SRR36875835.

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Conflicts of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest. All opinions expressed in this paper are the authors' and do not necessarily

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